

Zhuhan Bao

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Education

Hangzhou City University, B.Eng. in Software Engineering
GPA: 3.77/4.0 Average: 86.29/100

Sep 2023 – Jul 2027

Publications

SIMAX: A Scalable and Interpretable Framework for Multi-fidelity and Annotated Clinical Dialogue Simulation

Zhuhan Bao[†], Rui Yang[†], Bohao Yang, et al., Chuan Hong^{*}

Submitted to npj Digital Medicine, in editorial review

■ arXiv:2606.30491 🔗 AutoEvaluation/SIMAX

Co-first author; built the end-to-end multimodal clinical dialogue simulation pipeline and led its evaluation.

Toward Realistic Evaluation of Clinical Diagnostic Reasoning through Multimodal Evidence Synthesis

Rui Yang, Weihao Xuan, Zhuhan Bao, Huitao Li, et al., Nan Liu^{*}, Yifan Peng^{*}

Working Paper (not publicly indexed)

Developed ClinMM-Bench and conducted atomic-fact-based evaluation of multimodal clinical reasoning.

Research Experience

Visiting Scholar, Advisor: Prof. Nan Liu

Sep 2026 – Dec 2026

Centre for Biomedical Data Science

Duke-NUS Medical School, Singapore

Conduct research on medical AI, including multilingual clinical dataset generation and multimodal diagnostic reasoning evaluation.

Visiting Student, Advisor: Prof. Chuan Hong

Jul 2025 – Present

Department of Biostatistics & Bioinformatics, Duke University School of Medicine

Duke University, Durham, NC, USA

Develop multimodal clinical dialogue simulation and synthetic healthcare data systems; also maintain cluster research environments and benchmark H200 workloads.

Visiting Student, Advisor: Prof. Peiyu Liu

May 2025 – Aug 2025

Zhejiang University NESA Research Lab

Zhejiang University, Hangzhou, Zhejiang, China

Investigated MCP server security in AI code editors and built 🔗 MCP-Collector for automated marketplace data collection and classification.

Undergraduate Researcher, Advisor: Prof. Lin Sun

Mar 2025 – Mar 2026

School of Computer and Computing Science

Hangzhou City University, Hangzhou, Zhejiang, China

Led construction of a bilingual pathology image-text dataset and LoRA fine-tuning of CLIP for medical retrieval and diagnosis.

Teaching Experience

Teaching Assistant, Advisor: Prof. Kui Su

Spring 2026

High Performance Computing — Designed and graded assignments on MPI, OpenMP, and parallel algorithms.

Project Experience

Embodied World Model Inference Optimization

Jan 2026 – Mar 2026

2026 ASC Student Supercomputer Challenge

Accelerated UnifoLM-WMA-0 inference by $2.68\times$ using mixed precision, SDPA, CUDA Graphs, fused kernels, and an asynchronous data pipeline while preserving PSNR ≥ 25 dB.

RNA 5-Methylcytosine Construction Optimization

Jan 2025 – Feb 2025

2025 ASC Student Supercomputer Challenge

Built and optimized a reproducible RNA m5C detection pipeline covering cleaning, alignment, deduplication, and statistical filtering.

Learning Experience

Competitions

1st National Service Computing Innovation Competition

Excellent Award

2025 ASC Student Supercomputer Challenge

Second Prize

2026 ASC Student Supercomputer Challenge

Second Prize

16th Lanqiao Cup C/C++ Programming, Zhejiang Province

Second Prize

Captain of School High-Performance Computing Lab

Mar 2025 – Mar 2026

Hangzhou City University Supercomputing Center

Captain of the Voluntary Maintenance Team

Sep 2024 – Sep 2025

Voluntary Maintenance Team of the CS School

Skills

Certificates: CET-4, CET-6, PRC Driver's License (C1)

Programming Languages: C/C++, MPI, Python, Bash, Triton, CUDA

Skills: Inference Optimization, Parallel Computing, Machine Learning, Data Analysis, Algorithm Design

Software & Platforms: Linux, Slurm, Intel VTune, perf, SSH, Docker, Git, Conda, PyTorch